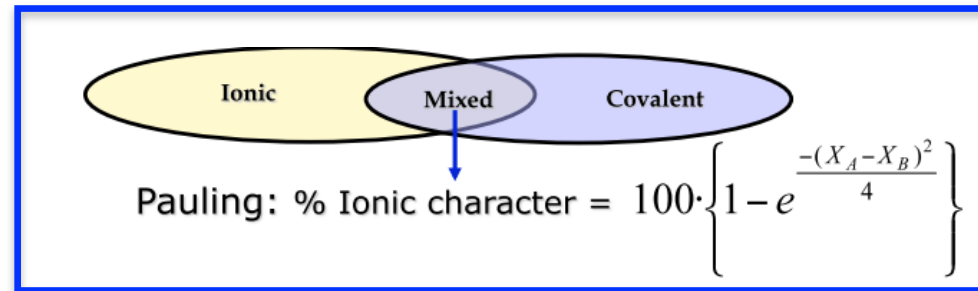


What are *ceramics*?

- The word *ceramic* was initially derived from the Greek word *keramos*, which means roughly “Potters clay”
 - Broad definition: “the art and science of making and using solid articles formed by the action of heat on earthy raw materials”
 - W. David Kingery: “the art and science of making and using solid articles which have as their essential component, and are composed in large part of, inorganic nonmetallic materials” - high melting points and thus require high temperature for processing.
 - Ceramics contain two types of primary bonds - *ionic and covalent*.
 - Ionic compounds generally form between very active metallic elements and active nonmetals.
 - Thus, the metals mostly involved in the formation of ceramics by ionic bonding are those from groups *IA*, *IIA*, and part of *IIIA*, as well as some of the transition metals.
 - Moreover, the most active nonmetals of groups *VII A* and *VIA* form ionic bonding.
- For covalent bonding to occur, ionic bonding should be unfavorable to form.
- If two elements forming a bond have similar electronegativities (Δx becoming very small).

- In ceramics, which bond predominates can be predicted from the value of Δx .
- A bond will be predominantly ionic when $\Delta x > 1.7$ and predominantly covalent if $\Delta x < 1.7$.
- Cs (0.79) and F (3.98), we have the Δx value between these two elements equal to 3.19, which is much above 1.7.
- Therefore, CsF is a predominantly ionic solid.



Percentage of ionic and covalent character of the bond for some ceramic materials ➡ **determines the CRYSTALLINE STRUCTURE**

Ceramic Material	Atoms in bond	$X_A - X_B$	% Ionic Character	% Covalent Character
MgO	Mg—O	2,3	73	27
Al ₂ O ₃	Al—O	2,0	63	37
SiO ₂	Si—O	1,7	51	49
Si ₃ N ₄	Si—N	1,2	30	70
SiC	Si—C	0,7	11	89

<i>Property</i>	<i>Ceramic</i>	<i>Metal</i>	<i>Polymer</i>
<i>Density</i>	Low	High	Very Low
<i>Hardness</i>	Very high	Low	Very low
<i>Wear resistance</i>	High	low	low
<i>Ductility</i>	Low	High	High
<i>Corrosion resistance</i>	High	Low	Low
<i>Thermal conductivity</i>	Mostly low	High	Low
<i>Electrical conductivity</i>	Mostly low	High	Low

low weight applications

wear resistant applications

difficult to give shape

corrosion resistant applications

can be used at high temp

but brittle in nature

type of bond : depends on delta x
 > 1.7 : ionic
 < 1.7 : covalent

Characteristics of Ceramics

- Brittle
- Chemically inert
- Long life
- High hard and wear resistant, electrically and thermally insulating, refractory, non-magnetic.

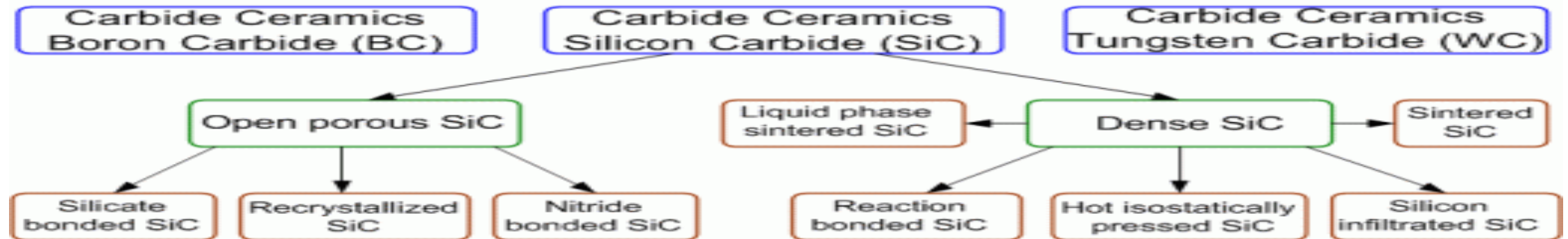
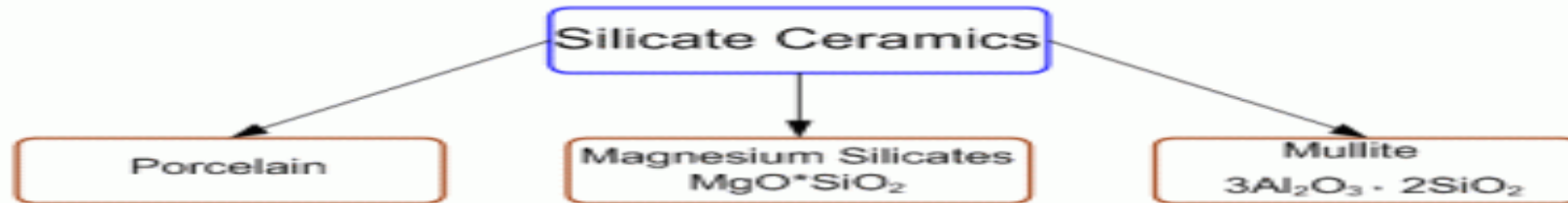
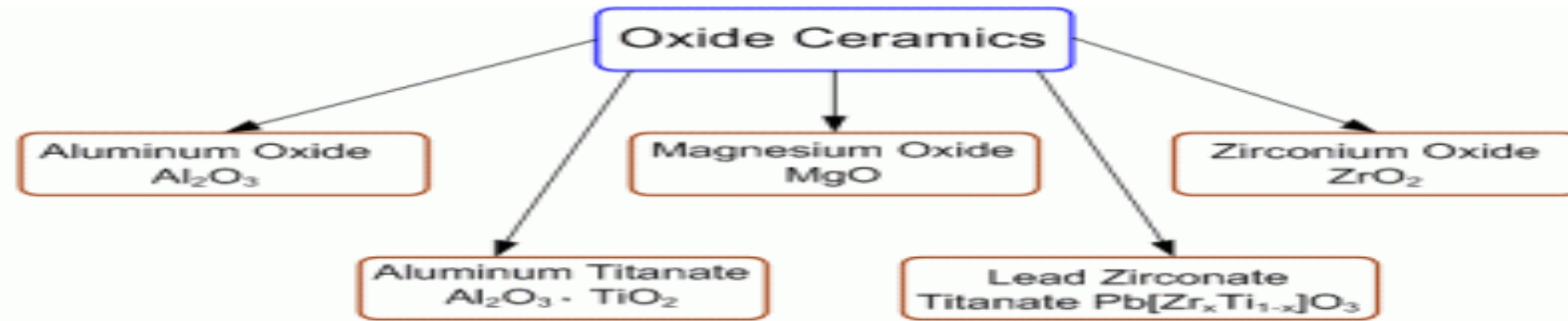
These properties are however not common to all ceramics

Some exceptions

- $\text{YBa}_2\text{Cu}_3\text{O}_7$ (Superconductors)
- $(\text{BaSr})\text{O}_{0.6}\text{Fe}_2\text{O}_3$ (Hard magnets)
- Partially stabilized zirconia (High toughness)
- New high performance ceramics
- Unusual properties (high toughness, high conductivity)

TABLE 14-3 ■ Mechanical properties of selected advanced ceramics

Material	Density (g/cm³)	Tensile Strength (psi)	Flexural Strength (psi)	Compressive Strength (psi)	Young's Modulus (psi)	Fracture Toughness (psi √in.)
Al ₂ O ₃	3.98	30,000	80,000	400,000	56×10^6	5,000
SiC (sintered)	3.1	25,000	80,000	560,000	60×10^6	4,000
Si ₃ N ₄ (reaction bonded)	2.5	20,000	35,000	150,000	30×10^6	3,000
Si ₃ N ₄ (hot pressed)	3.2	80,000	130,000	500,000	45×10^6	5,000
Sialon	3.24	60,000	140,000	500,000	45×10^6	9,000
ZrO ₂ (partially stabilized)	5.8	65,000	100,000	270,000	30×10^6	10,000
ZrO ₂ (transformation toughened)	5.8	50,000	115,000	250,000	29×10^6	11,000



Engineering ceramics can be classified into three - *oxides, nonoxides, and composites*.

➤ Examples for *oxides are alumina (Al_2O_3) and zirconia (ZrO_2).*

➤ *Carbides, borides, nitrides, and silicides come under nonoxides: SiC , Si_3N_4 .*

➤ Particulate-reinforced oxides and nonoxides are examples for composites

➤ *Oxide ceramics* are characterized by oxidation resistance, chemical inertness, electrical insulation and low thermal conductivity.

➤ Low oxidation resistance, extreme hardness, chemical inertness, high thermal conductivity, and electrical conductivity are the properties of *nonoxide ceramics*.

➤ *Ceramic-based composites* possess toughness, low and high oxidation resistance (depends upon the type), and variable thermal and electrical conductivity. The manufacturing processes are complex, and finally result in high cost

Classification based on composition

Nitride ceramics: nitrogen atoms bonded to elements such as silicon and aluminium

Silicon nitride (Si_3N_4) : White, , high melting point, chemically inert attacked by dilute HF and hot H_2SO_4

Good shock resistance compared to other ceramics (ball bearings) 80% less friction, 3 to 10 times longer lifetime

Better high temperature capability : creep and oxidation resistance.

Low thermal expansion coefficient

Insulator and chemical barrier in manufacturing integrated circuits

Orthopedic applications : alternative to PEEK and titanium for spinal fusion devices

Metal working and cutting tools : hardness, thermal stability and resistance to wear.

Aluminium nitride (AlN): Covalent bonded ceramic, mostly stable at high temperature in inert atmosphere

Air: surface oxidation occurs at 700°C , 5-10nm film thickness; protects the layer upto 1370°C ; bulk oxidation occurs,

III-V semiconductor group, Wurtzite crystal structure

High thermal conductivity with excellent electrical insulator prop.

Application: optoelectronics, electronic substrate as a crucible to grow crystals of gallium arsenide, Power and microelectronics, heat sink material)

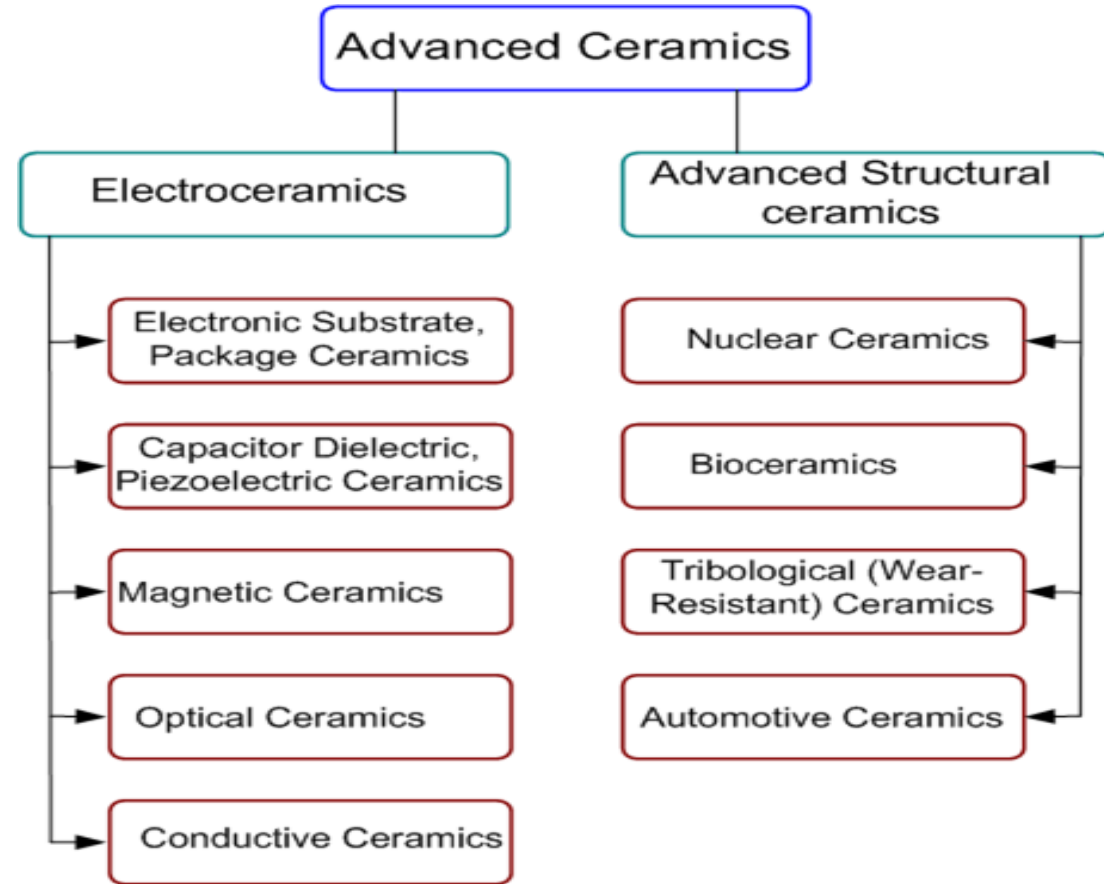
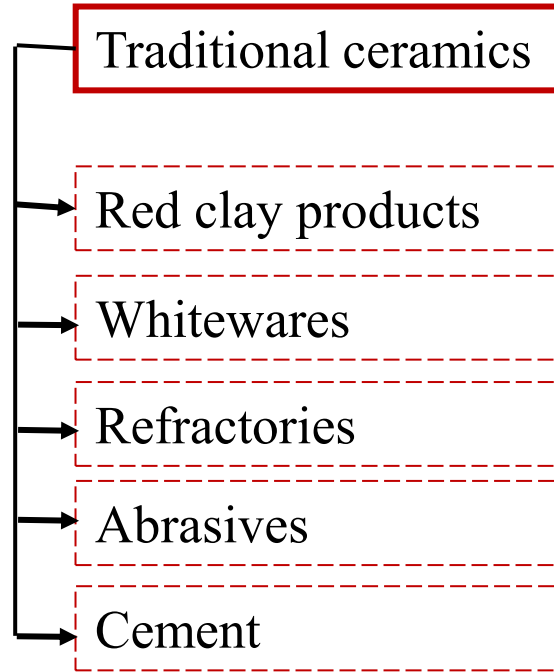
oxynitride

High temperature refractory materials, with high strength, good thermal shock resistance

Solid solution of nitrides and oxides

Silicon oxynitride, SiO_xN_y , have been of interest both as a higher dielectric constant gate and as an oxidation mask and diffusion barrier. It can be formed both by thermal nitridation of silicon and by CVD methods.

Classification based on applications



Traditional ceramics

➤ Red clay products : Stoneware, earthenware, Building blocks

Earthenware: clay fired at relatively low temperatures (1800°F-2100°F), often contains iron and has a porous surface when fired

Stoneware: a buff, gray or brown clay which is heavy, opaque, and highly plastic

in nature with a high firing temperature (2200°F-2400°F)

➤ *White clay products:*

Crockery, tiles, sanitary ware, electrical insulators



➤ **Refractories:** Fire clay, high alumina, magnesite

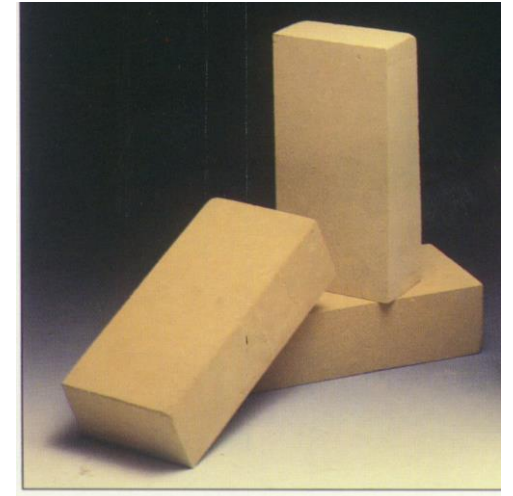
Firebricks for furnaces and ovens. Have high Silicon or Aluminium oxide content.

Brick products are used in the manufacturing plant for iron and steel, non-ferrous metals, glass, cements, ceramics, energy conversion, petroleum, and chemical industries

Used to provide thermal protection of other materials in very high temperature applications, such as steel making ($T_m=1500^\circ\text{C}$), metal foundry operations,

They are usually composed of alumina ($T_m=2050^\circ\text{C}$) and silica along with other oxides: MgO ($T_m=2850^\circ\text{C}$), Fe_2O_3 , TiO_2 , etc.,

Specialized refractories, (those already mentioned) and BeO , ZrO_2 , mullite, SiC , and graphite with low porosity are also used.



Raw materials

Traditional ceramics

1) Plastic material

Assist forming process (deform easily without rupture, retain the imposed shape)

Example: Clays, Talc

The clay minerals are hydrated aluminosilicates that have layer structures.

There are a variety of clay minerals; including kaolinite ($\text{Al}_2(\text{Si}_2\text{O}_5)(\text{OH})_4$), halloysite ($\text{Al}_2(\text{Si}_2\text{O}_5)(\text{OH})_4 \cdot 2\text{H}_2\text{O}$), pyrophyllite ($\text{Al}_2(\text{Si}_2\text{O}_5)_2(\text{OH})_2$), and montmorillonite ($\text{Al}_{1.67}(\text{Na},\text{Mg})_{0.33}(\text{Si}_2\text{O}_5)_2(\text{OH})_2$).

2) Flux:

Melts at relatively low temperature, reacts with other components, producing a viscous liquid, which acts as hard binder on cooling

Example: Feldspar; is also used in glass, pottery, enamel, and other ceramic products. Feldspar minerals

range in composition from KAlSi_3O_8 to $\text{NaAlSi}_3\text{O}_8$ to $\text{CaAl}_2\text{Si}_2\text{O}_8$ and act as a flux (reduce the melting temperature) in a composition. Nepheline syenite ($\text{Na}_2\text{Al}_2\text{Si}_2\text{O}_8$) is used in a similar fashion.

3) Filler

Provides rigid component to aid in forming and firing. Example: Silica (SiO_2)

Functional ceramics

Functionality	Material	Applications
Resistors	SiC, MoSi ₂ , LaCrO ₃	Heating elements for high temperature furnaces
Thermistors (NTCR & PTCR)	Spinel BaTiO ₃	Temperature sensors, self-regulating heating elements
Dielectrics with very low losses ($\epsilon_r = 3 - 10$)	Al ₂ O ₃ , AlN, cordierite	Substrates for electronic circuits and chip packaging
Dielectrics for microwave applications ($\epsilon_r = 30 - 80$)	BaTi ₄ O ₉ , Zr(Ti,Sn)O ₄ , BaMg _{1/3} Ta _{2/3} O ₃ , (Ba,Sr)TiO ₃ ,	MW resonators, filters and antennas for mobile communications and GPS devices, tunable MW devices
Temperature stable dielectrics ($\epsilon_r \approx 100$)	CaTiO ₃ , BaO-Nd ₂ O ₃ -TiO ₂	Capacitors with temperature-independent capacitance
Dielectrics with very high dielectric constant ($\epsilon_r \approx 3000$)	BaTiO ₃	Multilayer ceramic capacitors
Piezoelectric ceramics	Pb(Zr,Ti)O ₃ (PZT)	Transducers, actuators and resonators
Pyroelectric ceramics	Pb(Zr,Ti)O ₃	IR radiation detection and imaging

Functional ceramics

Functionality	Material	Applications
Ferroelectric ceramics	$\text{Pb}(\text{Zr},\text{Ti})\text{O}_3$ $\text{SrBi}_2\text{Ta}_2\text{O}_9$	Ferroelectric memories (FeRAMs)
Electrostrictive ceramics	$\text{PbMg}_{1/3}\text{Nb}_{2/3}\text{O}_3$ - PbTiO_3 (PMN-PT)	Actuators
Magnetic ceramics	Spinel $(\text{Ni},\text{Zn})\text{Fe}_2\text{O}_4$ $\text{BaFe}_{12}\text{O}_{19}$ $\text{Y}_3\text{Fe}_5\text{O}_{12}$ (YIG)	Inductors Permanent magnets Microwave devices (radars)
Ionic conductors	Y:ZrO_2 (YSZ) Gd:CeO_2 β -alumina	Electrolytes for solid-oxide fuel cells (SOFCs), oxygen sensors Na-Batteries
Superconductors	$\text{YBa}_2\text{Cu}_3\text{O}_{7-x}$ (YBCO) MgB_2	Superconducting cables for magnets
Transparent ceramics	Al_2O_3 , MgAl_2O_4 , $\text{Y}_3\text{Al}_5\text{O}_{12}$ (YAG)	Phosphors, optical materials for lenses and laser systems, nose cones for heat-seeking missiles, high-pressure sodium street lamps
Optoelectronic materials	LiNbO_3 PLZT	Waveguides, frequency doublers, voltage-controlled optical switches, modulators

Structural ceramics



Two Kyocera ceramic knives
(Y:ZrO_2)



Ceramic body armour plates
(Al_2O_3 , SiC)



Ceramic Si_3N_4 bearing parts



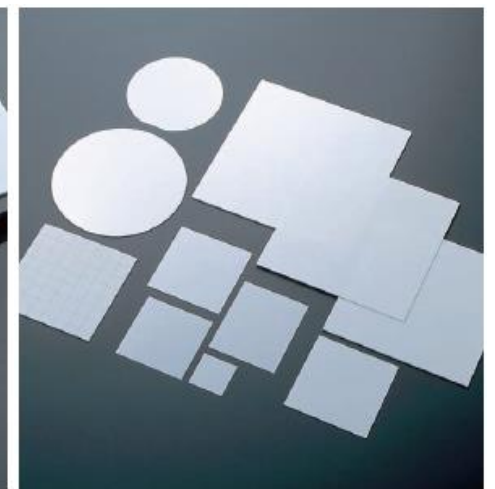
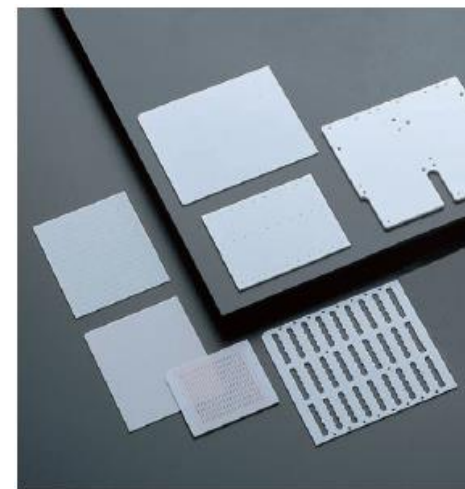
Radial rotor made from Si_3N_4 for
a gas turbine engine



The Porsche Carrera GT's
silicon carbide disk brake



Pressed and extruded parts (alumina, mullite,
zirconia)



Thick (left) and thin (right) substrates (alumina)

MAKING POWDERS

Many methods are available for the preparation of ceramic powders.

These can be divided into just three basic types:

- ❑ **Mechanical, Chemical, Vapor phase**

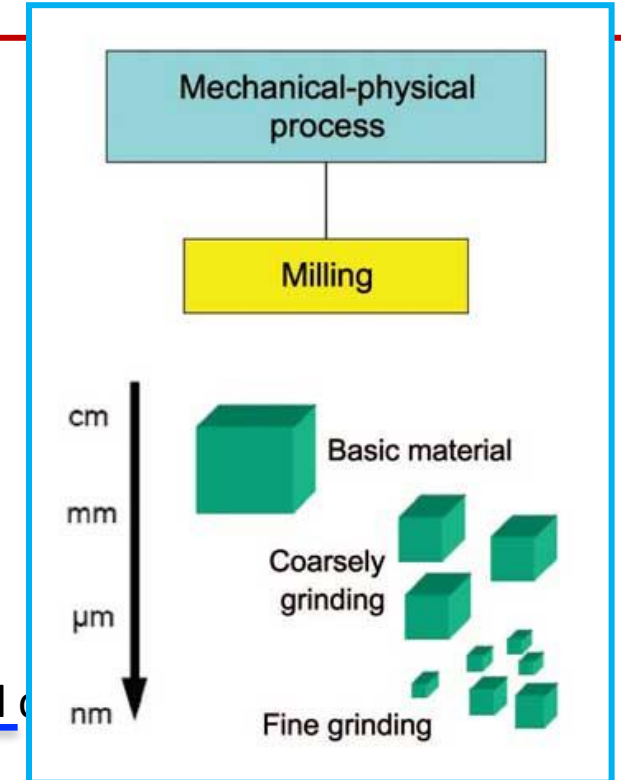
- ❑ **Mechanical methods** use coarse-grained materials that have generally been derived from naturally occurring minerals

- They are subjected to a series of processes in which the particle size is gradually reduced. The final step is known as milling, which produces particles of the desired size.
- Mechanical methods of powder production are used widely in the production of traditional purity powders are not required and cost is one of the most important requirements

- ❑ **Chemical methods**, such as sol-gel processing, offer several advantages over mechanical methods because they allow exceptional control over particle morphology and purity. Chemical processes are used widely in the production of advanced ceramic materials.

- ❑ **Vapor-phase processes can be used to produce ceramic powders.**

They tend to be expensive, but offer many advantages, such as the ability to produce particles of nonoxides.



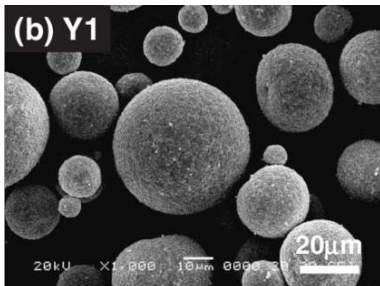
Desirable Powder Characteristics for Advanced Ceramics

<i>Powder characteristic</i>	<i>Desired property</i>
Particle size	Fine ($<1\text{ }\mu\text{m}$)
Particle size distribution	Narrow
Particle shape	Spherical or equiaxed
State of agglomeration	No agglomeration or soft agglomerates
Chemical composition	High purity
Phase composition	Single phase

TYPES OF POWDERS

Primary particles are the smallest clearly identifiable unit in the powder.

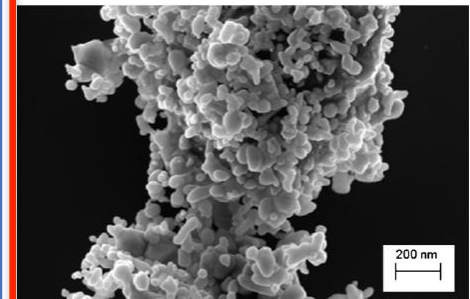
Primary particles may be crystalline or amorphous and cannot easily be broken down into smaller units.



Agglomerates are clusters of bonded primary particles.

Soft agglomerates are easily broken up; hard agglomerates, because of the stronger interparticle bonds, are more difficult to break up.

Hard agglomerates should be avoided in ceramic powder processing as much as possible.



Particles is a general term applied to both primary particles and agglomerates.

Granules are large agglomerates, usually 0.1–1 mm in diameter, that are formed by the addition of a granulating agent (e.g., a polymer binder).

The mixture is tumbled, producing large, nearly spherical granules that flow freely and can be used to fill complex molds.

Flocs are clusters of particles in a liquid suspension held together electrostatically.

Colloids are very fine particles (they can be as small as 1 nm in diameter) held in fluid suspension by Brownian motion. Consequently, colloidal particles will settle very slowly.

Aggregates are coarse constituents, >1 mm, in a mixture. The important example is the addition of gravel to cement to make concrete. In early concrete structures such as the Pantheon in Rome, pumice was used as aggregate.

Carbonates, nitrates, sulfates, oxalates, and other compounds containing oxygen ions are commonly used in preparing ceramic powder batches

Modern ceramics preparation techniques

Pure oxides: Chemical purification, Sol-gel synthesis, Hydrothermal, Spray Pyrolysis, CVD

Mixed Oxides: Solid state synthesis, Co-Precipitation, Sol-gel synthesis, Hydrothermal

Non-oxide compounds: Carbothermal, solid-gas reaction, Solid state reaction

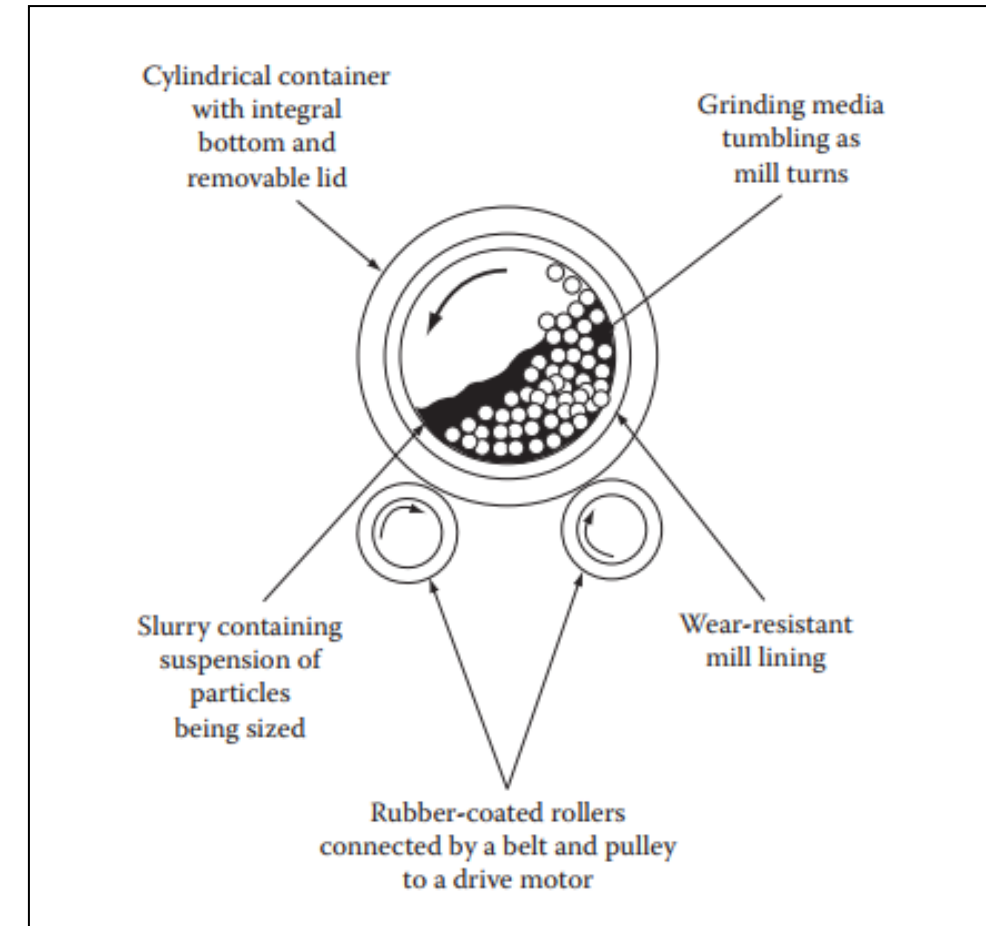
Processing of ceramics

- ☐ Batch preparation of powder raw materials
- ☐ Consolidation
- ☐ Firing/sintering
- ☐ finishing

MECHANICAL MILLING



- For traditional raw materials like clay and the oxides produced from ores, it is often necessary to eliminate aggregates and to reduce the particle size.
- The most common method for reducing particle size is ball milling.
- A ball mill is a barrel (usually made of a ceramic,) that rotates on its axis and is partially filled with a grinding medium (called media) in the form of spheres, cylinders, or rods
- The quantity of the media is such that the rotation of the mill causes it to cascade, creating both shearing and crushing actions on the powder.



Process variables

- *Types of mill*
- *Milling container*
- *Milling speed*
- *Milling time*
- *Grinding medium*
- *Ball to powder ratio*
- *Milling atmosphere*
- *Process control agents*

Possible Particles Sizes for Different Milling Techniques

Jaw crushers	to 5 mm
Cone crushers	to 5 mm
Crushing rolls	to ~1 mm
Hammer mill	to ~0.1 mm
Jet mill	1 to ~50 μm
Vibratory mill	1 to ~50 μm
Ball mill	0.5–10 μm
Attrition mill	0.1–5 μm
Roller mill	0.1–5 μm

The advantages of ball milling are that the equipment is

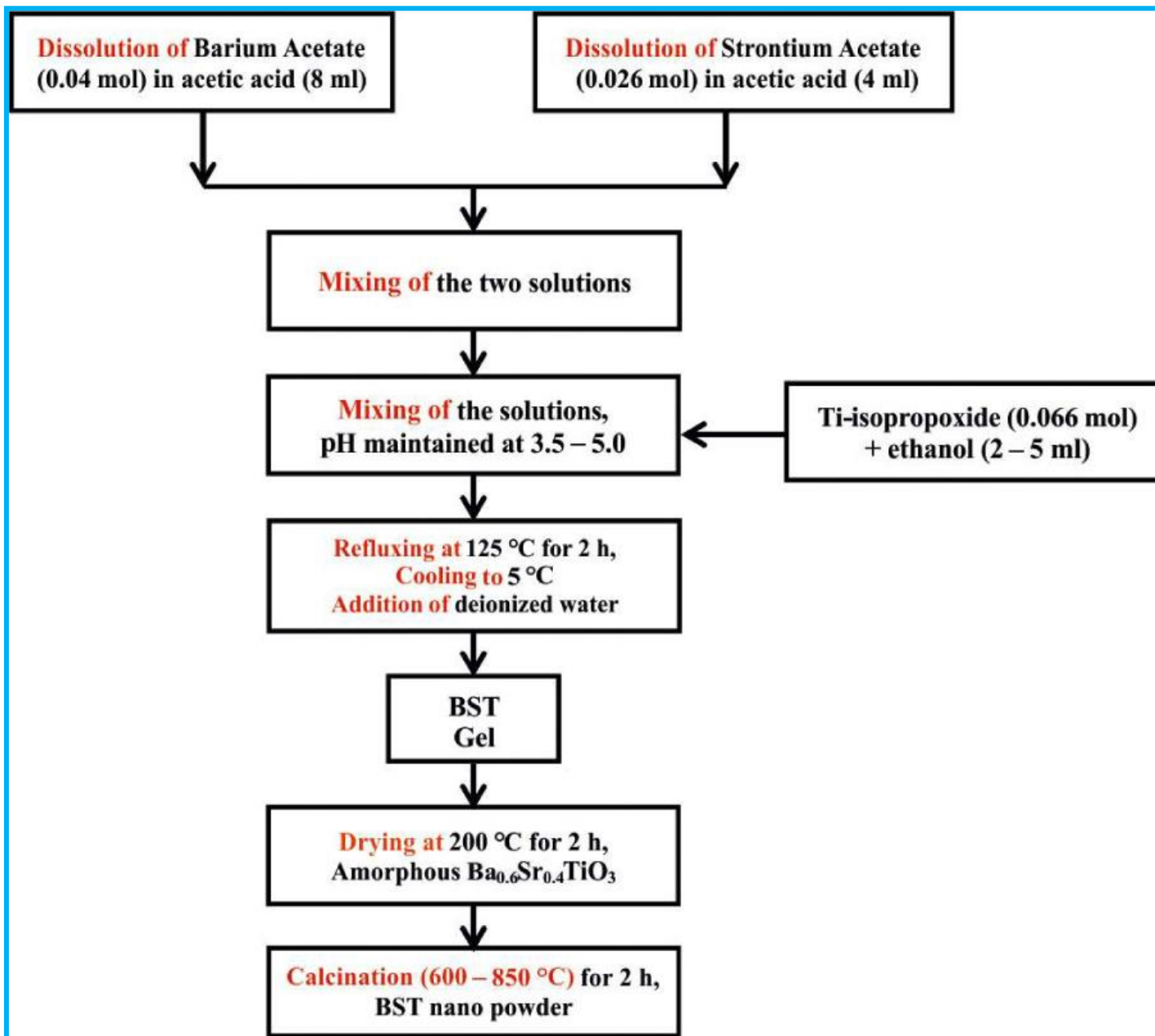
Simple (although experimentally straightforward, there are many theoretical aspects that are quite complex)

Inexpensive (at least for small batch sizes)

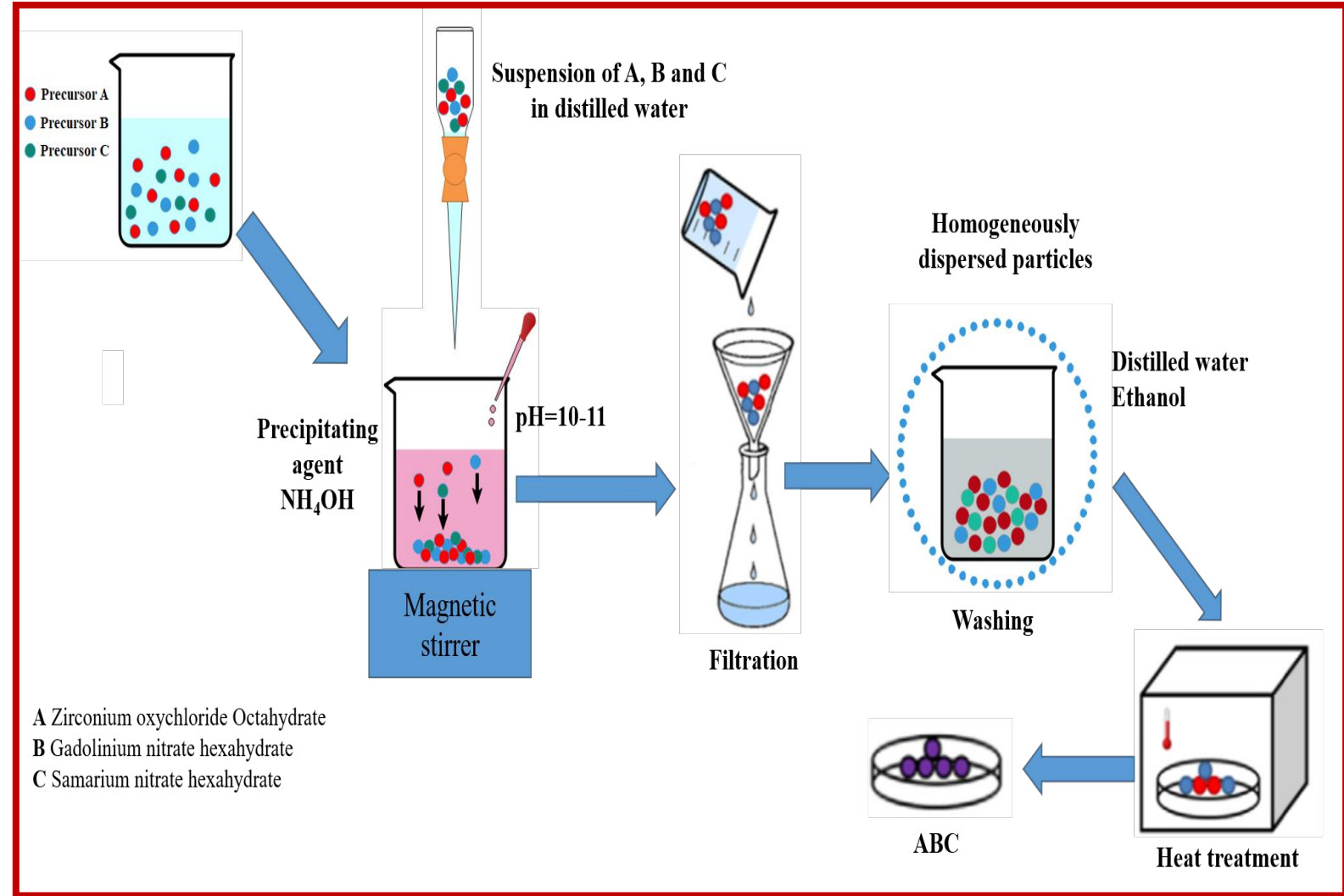
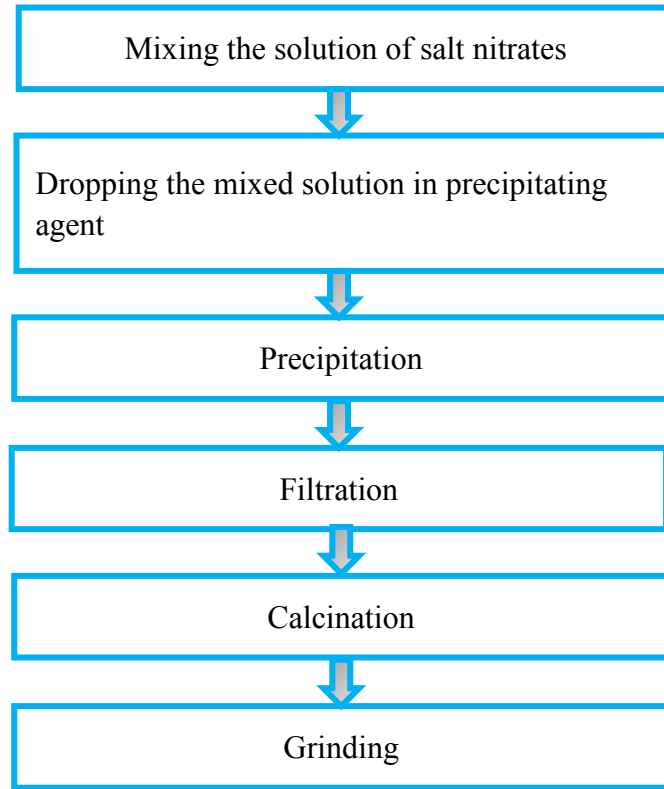
Sol–Gel Processing of Ceramics

- *Sol–gel* is a generic term that includes a variety of techniques to achieve a high-purity composition with homogeneity at the molecular level
- based on solution-based processing, have been developed to gain a more complete control over the structure of the nanometer-sized building blocks, and to allow the synthesis of ceramics at lower processing temperatures
- Typical applications of sol–gel materials include all types of functional coatings (scratch-resistant, barrier properties etc.), catalysts and catalyst supports, ceramic fibers, electroceramic powders (energy storage and conversion), insulating materials, high-purity glasses, biomaterials, among others.
- The sol–gel process is a wet chemical process that can be described as the creation of a network by progressive polycondensation reactions of molecular precursors in a liquid medium.
- Alternately, it is a process used to form materials via a sol, followed by gelation of the sol and finally removal of the solvent, typically resulting in materials that comprise nanoscale features.
- As a solution-based approach, sol–gel processing allows the fabrication of materials with a defined pore design, pore size distribution, pore interconnectivity and network on various length scales by adjusting the type of solvent, catalyst, temperature and the presence or absence of templating molecules

- The starting materials to synthesize $\text{BaSr}_{0.5}\text{Ti}_{0.5}\text{O}_3$, barium acetate $[\text{Ba}(\text{CH}_3\text{COO})_2]$; strontium acetate $[\text{Sr}(\text{CH}_3\text{COO})_2]$; and titanium tetra isopropoxide (TTIP) as barium, strontium and titanium precursors, respectively.
- Glacial acetic acid and ethanol were used as the solvent while acetylacetone was used as a chelating agent.
- All of the starting materials used in the experiment were of analytical grade reagents.
- For the preparation of BS sols, the barium and strontium acetates were initially dissolved in acetic acid with molar ratio of 1:1 and stirred at 40 °C for 0.5 h until becoming a transparent and clear solution.
- Afterward, TTIP was dissolved in absolute ethanol and then acetyl acetone was added to it. In the next step, this solution was added to (Ba, Sr) precursor solution and stirred to form a sol, which contained barium, strontium and titanium.
- The prepared sols were dried in a muffle furnace at 100 °C for 5 h to be dehydrated and then were heat-treated up to 650, 750 and 850 °C at rate of 10 °C/min and held for 2 h to obtain the crystalline powders.
- The calcined powders were isostatically pressed into pellets 10 mm in diameter and average thickness of about 3 mm using a hydraulic press at 100 MPa. The samples were placed into the tube furnace and sintered at temperature of 1300 °C for 2 h at the heating rate of 10 °C/min in an air atmosphere.



Chemical Coprecipitation-Calcination Method



Chemical coprecipitation-calcination method is a technique for removing metal ions from solution, where the metal ions are coprecipitated usually as hydroxides, carbonates.

- Spray drying is an example of powder production from solution. It is used widely for preparing ferrites, titanates, and other electrical ceramics. Fine droplets produced by an atomizer are sprayed into a drying chamber and the powder is collected.
- There are different types of atomizers.
- One uses ultrasonic atomization in which the solution is passed over a rapidly vibrating piezoelectric membrane.
- Droplet sizes in the range of micrometer can be produced.
- In the drying chamber, the flow pattern of the hot air determines the completeness of moisture removal and the maximum temperature that the particles experience.
- Finally the particles are carried out of the chamber in the air stream and captured in a bag or another form of collector.
- The particles produced by spray drying are often agglomerated with a primary particle size less than 0.1 μm .

The variables in spray drying are

- Droplet size
- Solution concentration and composition
- Temperature and flow pattern of the air in the drying chamber
- Chamber design

➤ For small-scale laboratory experiments nitrates and acetates are often used because of their relatively low decomposition temperature. Chlorides and oxychlorides are frequently used in industrial spray-drying operations because of their high solubility in aqueous solutions. The capacities of industrial spray dryers are up to several hundred kilograms per hour. The spray drying process is not limited to aqueous solutions; for example, alcohol solutions of alkoxides can be used.

<i>Precursor</i>	<i>T (°C)</i>
Zn(NO ₃) ₂ ·6H ₂ O	360
Ni(NO ₃) ₂ ·6H ₂ O	400
Ni(CH ₃ COO) ₂ ·4H ₂ O	350
Fe(NO ₃) ₃ ·9H ₂ O	200
MgSO ₄	1000
Y ₂ (C ₂ O ₄) ₃ ·5H ₂ O	500

***Salt Precursors and Their
Decomposition
Temperatures in Air***

Modern ceramics

❑ Aluminum Oxide Powder

- ❑ occurs naturally as the mineral corundum, which is better known to most of us when it is in gem-quality crystals called ruby and sapphire.
- ❑ Ruby and sapphire are precious gems because of their chemical inertness and hardness.
- ❑ Al_2O_3 powder is produced in large quantity from the mineral bauxite by the Bayer process.
- ❑ Bauxite is primarily colloidal aluminum hydroxide intimately mixed with iron hydroxide and other impurities.
- ❑ The Bayer process involves the selective leaching of the alumina by caustic soda and precipitation of the purified aluminum hydroxide.
- ❑ The resulting fine-particle-size aluminum hydroxide can then be thermally converted to Al_2O_3 powder, which is used to manufacture polycrystalline Al_2O_3 -based ceramics.

Appln: porcelain, alumina laboratory ware, crucibles and metal casting molds, high-temperature cements, wear-resistant parts (sleeves, tiles, seals, etc.), sandblast nozzles, armor, medical components, abrasives, refractories, and a variety of other components.

Many hundreds of tons of alumina powder and alumina-based articles are produced each year. It has even been used to make extrusion dies for corn chips and mixing valves for water faucets.

Silicon Carbide Powder

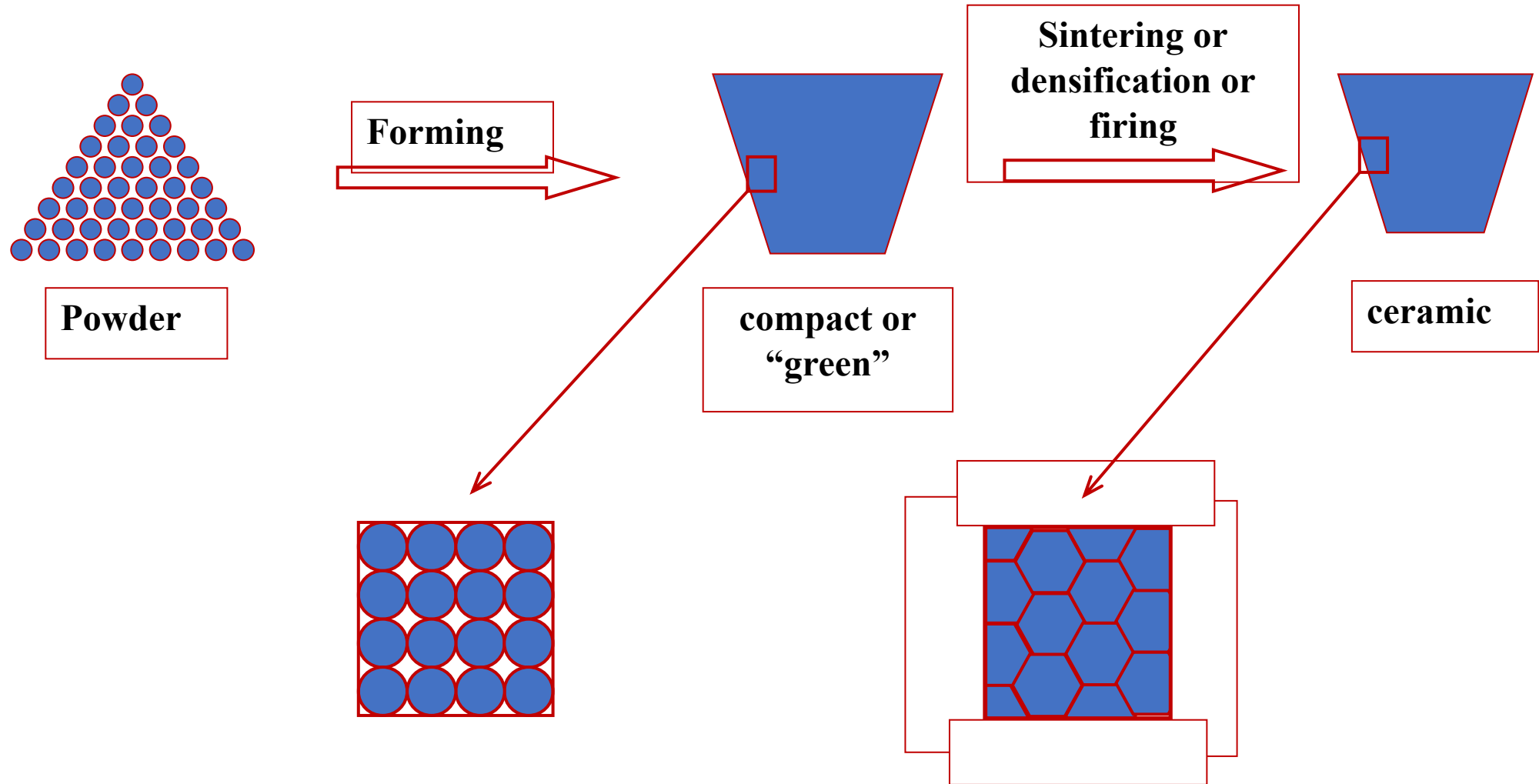
- ❑ SiC has been found occurring naturally only as small green hexagonal plates in meteoric iron. This same hexagonal polymorph (α -SiC) has been synthesized commercially in large quantities by the Acheson process. This fascinating process appears crude, but is cost-effective and simultaneously produces lower-grade SiC for abrasives and high-grade SiC for electrical applications.
- ❑ The Acheson process consists essentially of mixing SiO_2 sand with coke in a large elongated mound and placing large carbon electrodes in opposite ends. An electric current is then passed between the electrodes, resistance-heating the coke in the mound to about 2200°C . In this temperature range, the coke reacts exothermically with SiO_2 to generate enough heat to sustain a chemical reaction to produce SiC plus CO gas. Exothermic heating continues until the reaction is completed in the interior of the mound. After cooling, the mound is broken up and sorted.
- The core contains intergrown green hexagonal SiC crystals that are low in impurities and suitable for electronic applications.
- Around the core is a zone of lower purity that is used for abrasives. The outer layer of the mound is a mixture of SiC and unreacted SiO_2 and carbon that is added to silica sand and coke for the next batch.
- SiC can be prepared from almost any source of silicon and carbon.

- For instance, it has been prepared in the laboratory from a mixture of silicon metal powder and sugar. It has also been prepared from rice hulls.
- It can also be prepared from silicon tetrachloride (SiCl_4) and some silanes (SiH_4). In this case, the cubic β -SiC polymorph normally forms.

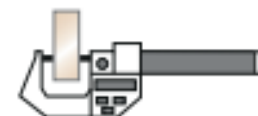
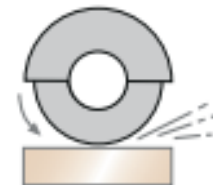
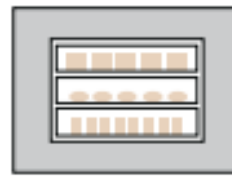
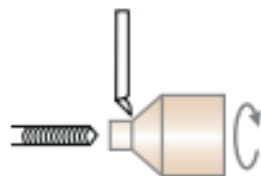
Appln: used for high-temperature kiln furniture, electrical resistance heating elements, grinding wheels and abrasives, wear resistance applications, incinerator linings, varistors, light-emitting diodes, and also gemstones cut from single crystals.

➤ Silicon Nitride Powder Si_3N_4

- Silicon nitride does not occur naturally, it has been synthesized by several different processes.
- Commercially, it has been made by the reaction of silicon powder with nitrogen at temperatures in the range of 1250–1400°C and consists of a mixture of $\alpha\text{-Si}_3\text{N}_4$ and $\beta\text{-Si}_3\text{N}_4$ polymorphs.
- This Si_3N_4 is not a ready-to-use powder when it is removed from the furnace. It is loosely bonded and must be crushed and sized first.
- The resulting powder is not of high purity, but contains impurities such as Fe, Ca, and Al, which were originally present in silicon, plus impurities picked up during crushing and grinding.
- Higher-purity Si_3N_4 powder has been made by reduction of SiO_2 with carbon in the appropriate nitrogen environment and reaction of SiCl_4 or silanes with ammonia.
- Both of these methods produce very-fine-particle-size powder that does not require further grinding for use.
- In fact, some of these powders are so fine that they require coarsening by heat-treating (calcining) before they are suitable for shape-forming operations.
- High-purity Si_3N_4 powder small quantities by laser reaction: A mixture of gaseous SiH_4 and ammonia is exposed to the coherent light output of a CO_2 laser. Silane has high absorption for the wavelengths involved, resulting in the heat required for reaction. spherical particles of a uniform size (20–100 nm) for the given gas flow and laser power conditions.



PRODUCTION PROCESS



Powder

Material preparation

Moulding

Green machining

Sintering

Finishing

Quality Assurance

▶ Raw materials

- ▶ Pulverizing
- ▶ Mixing
- ▶ Filtering
- ▶ Granulating
- ▶ Plasticizing
- ▶ Spray drying

- ▶ Dry pressing
- ▶ Extruding
- ▶ Injection moulding
- ▶ Isostatic pressing

- ▶ Milling
- ▶ Turning
- ▶ Drilling
- ▶ Cutting

- ▶ Debinding
- ▶ Sinter firing
900°to
1750°C

- ▶ Grinding
- ▶ Lapping
- ▶ Polishing
- ▶ Honing
- ▶ Metallize
- ▶ Glazing
- ▶ Assembly

- ▶ Testing
 - Optical
 - Dimensional check
 - Crack detection
 - Strength

Processing of ceramics

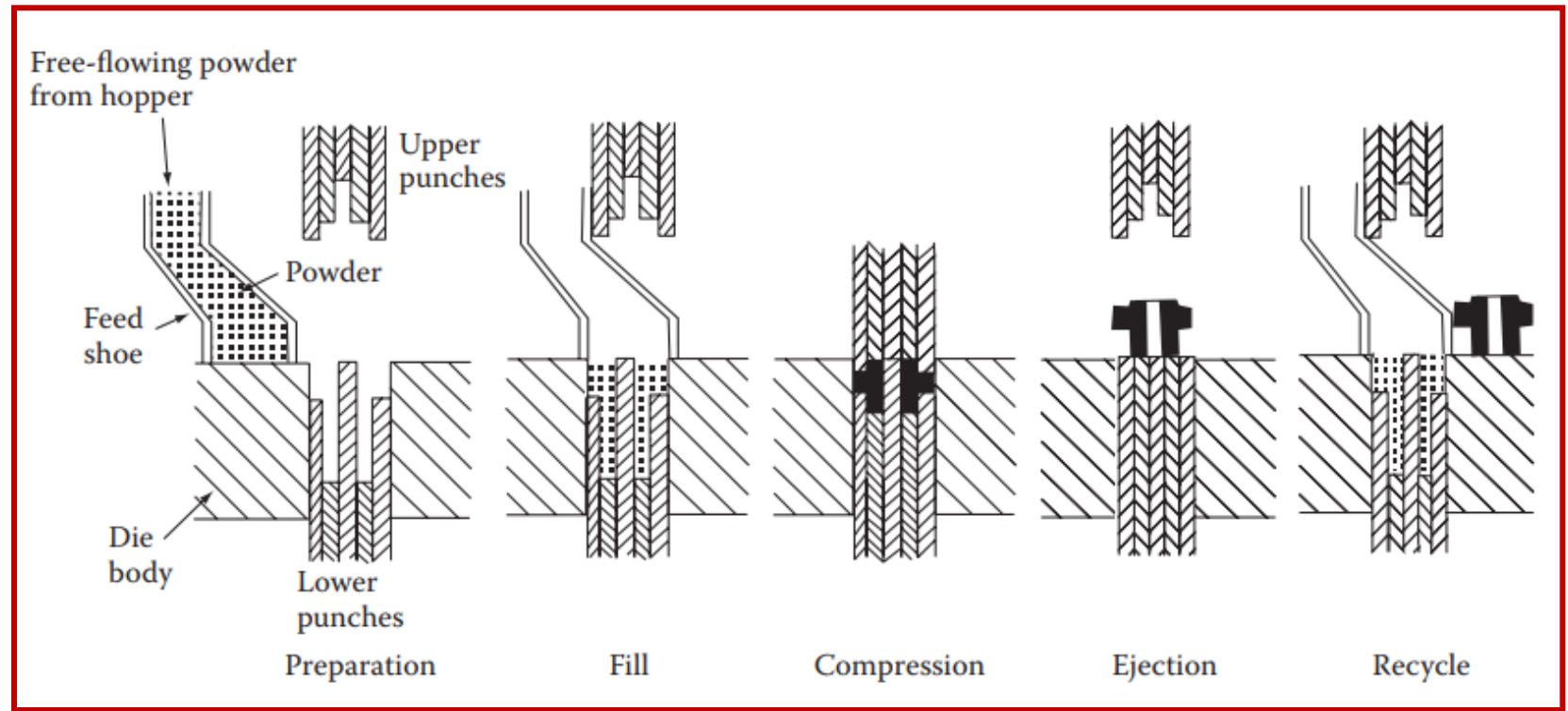
Shaping and Forming

1. *Powder compaction: e.g., dry pressing, hot pressing, isostatic pressing*
2. *Casting: using a mold with the ceramic as, or containing, a liquid or slurry*
3. *Plastic forming: e.g., extrusion, injection molding—using pressure to shape the green ceramic*

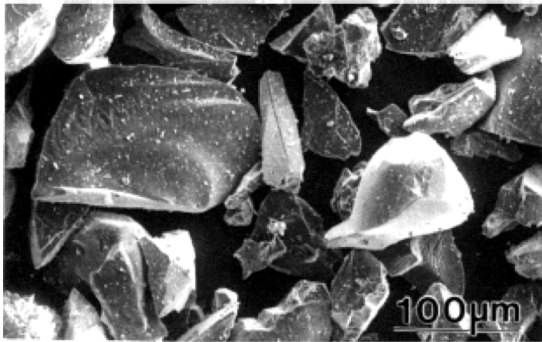
TABLE 23.1 Various Shaping Methods for Ceramic Components

<i>Shaping method</i>	<i>Type of feed material</i>	<i>Type of shape</i>
Dry pressing	Free-flowing granules	Small and simple
Isostatic pressing	Fragile granules	Larger and more intricate
Extrusion	Plastic mass using a viscous polymer solution	Elongated with constant cross section
Injection molding	Organic liquid binder giving fluidity when hot	Complex
Slip casting	Free-flowing cream	Mainly hollow

DRY PRESSING

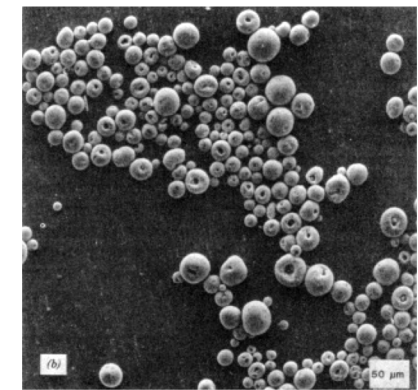


Particle packing and granulation



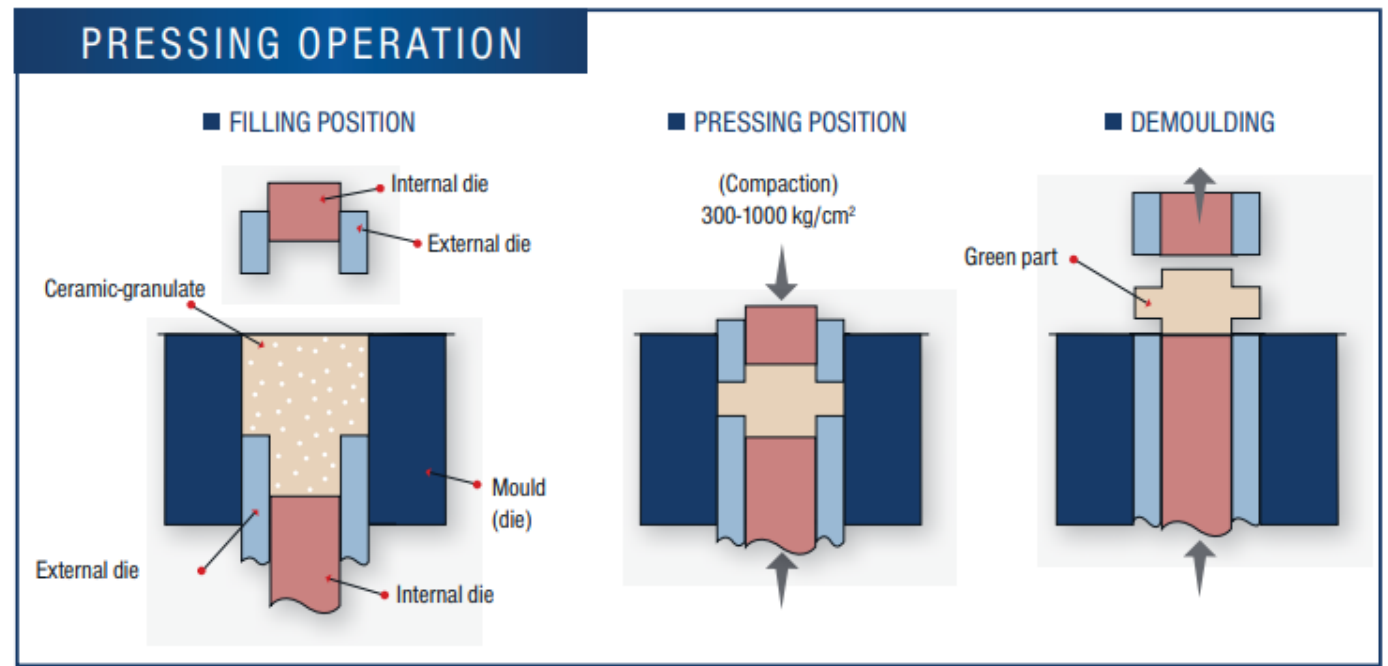
Angular shape of milled alumina

Particles with sharp edges or formed by irregular aggregates do not flow and do not pack efficiently. Granulation is essential.



Process

- Filling the die,
- Compacting the contents,
- Ejecting the pressed solid the powder
- mixture contains 0–5 wt% of a binder
- particle size distribution of 20–200 μm
- pressures used may be as high as 300 MPa



Advantages:

- ❑ Process is so simple and involves low-cost capital equipment
- ❑ High-volume forming process for ceramics.
- ❑ Production rates
- ❑ For large components (e.g., refractories) or complex parts (e.g., grinding wheels): 1–15 parts per minute.
- ❑ Cutting tools: several thousand per minute

Hot pressing

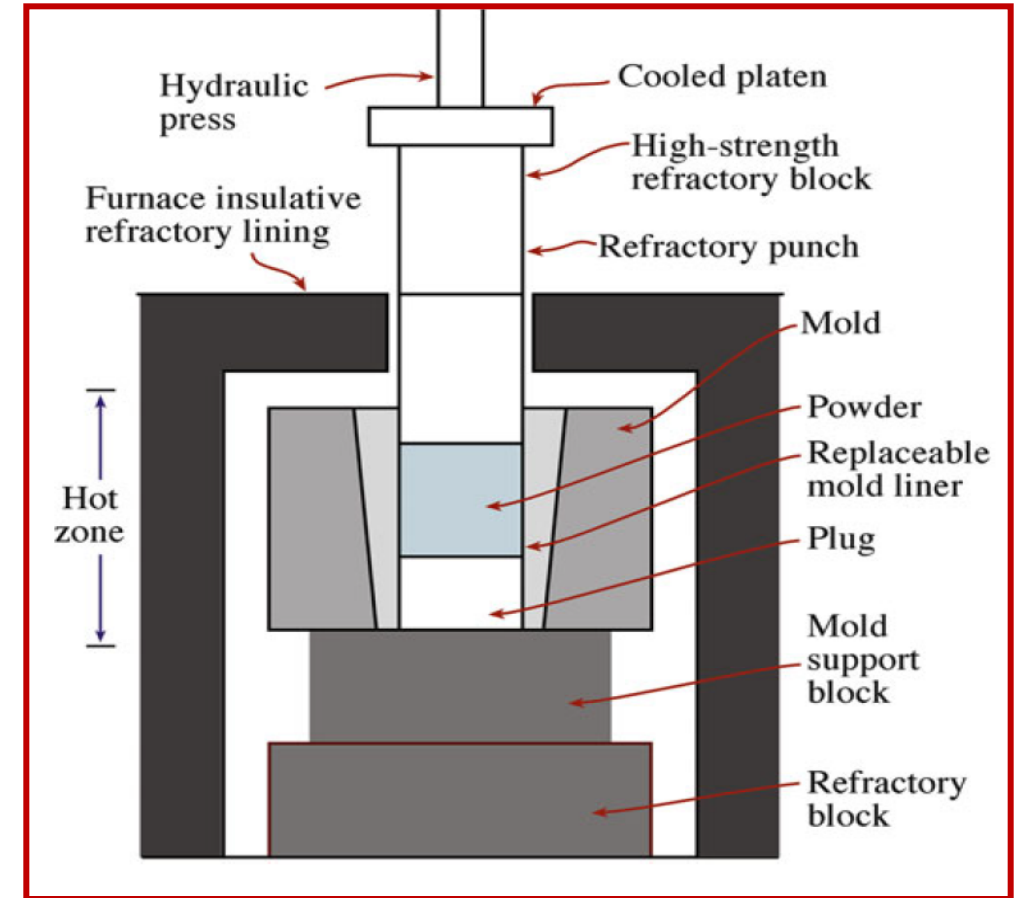
Advantages of Hot pressing

- ❑ The powder does not have to be of the highest quality.
- ❑ Large pores that are caused by non uniform mixing are easily removed.
- ❑ Densify at lower temperatures.
- ❑ Extensive grain growth does not occur when we keep the temperature low during densification.
- ❑ Covalently bonded materials such as B_4C , SiC , and Si_3N_4 without additives

Metals, alloys, mostly based on Mo ($1,000^{\circ}C$; 80 MPa)

Al_2O_3 , SiC , and Si_3N_4 - $1,400^{\circ}C$

Graphite - $2200^{\circ}C$; 10-30 MPa



Dies for use at high temperatures are expensive and generally don't last long.

Advantages of Hot pressing

- ❑ Limited to simple solid shapes, such as flat plates, blocks, and cylinders.
- ❑ More complex or large shapes are difficult and often impossible to produce by hot pressing.
- ❑ High cost and low productivity.

Hot Pressed Products

Ceramic armor	B_4C , TiB_2 , SiC , Al_2O_3
Cutting tools	Al_2O_3 particle-reinforced TiC , Si_3N_4 , $Si_3N_4-AlN-Al_2O_3$
Tooling (molds and dies)	$Al_2O_3-SiC_w$ composites, SiC , Al_2O_3
Sputtering targets	$Cr-SiO$, TiN , Si_3N_4 , B_4C , Al_2O_3
Heat engine components	Si_3N_4 , SiC
Ceramic bearings	Si_3N_4
Microwave absorbers	$MgO-SiC$ particulate composites, $BeO-SiC$, Al_2O_3-SiC
Varistors	ZnO
Electro-optic materials	PLZT
Titanates	$BaTiO_3$, $CaTiO_3$

ISOTATIC PRESSING:

Compaction takes place under isostatic or near-isostatic pressure conditions.

The green density of the isostatically pressed part is higher and much more uniform.

COLD ISOSTATIC PRESSING

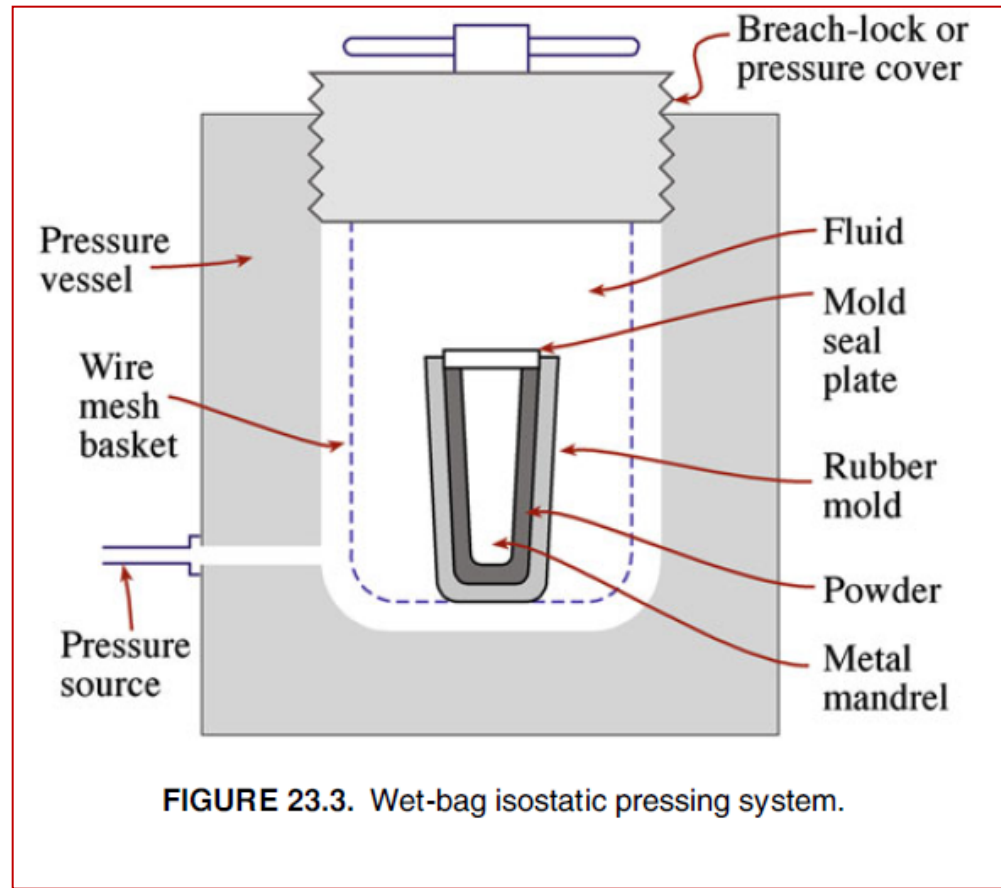
wet-bag isostatic pressing:

Pressures up to 400MPa to 1GPa.

Product: Tubes, crucibles.

Advantages

- Wide range of shapes and sizes that can be produced
- Uniform density of pressed product
- Low tooling costs



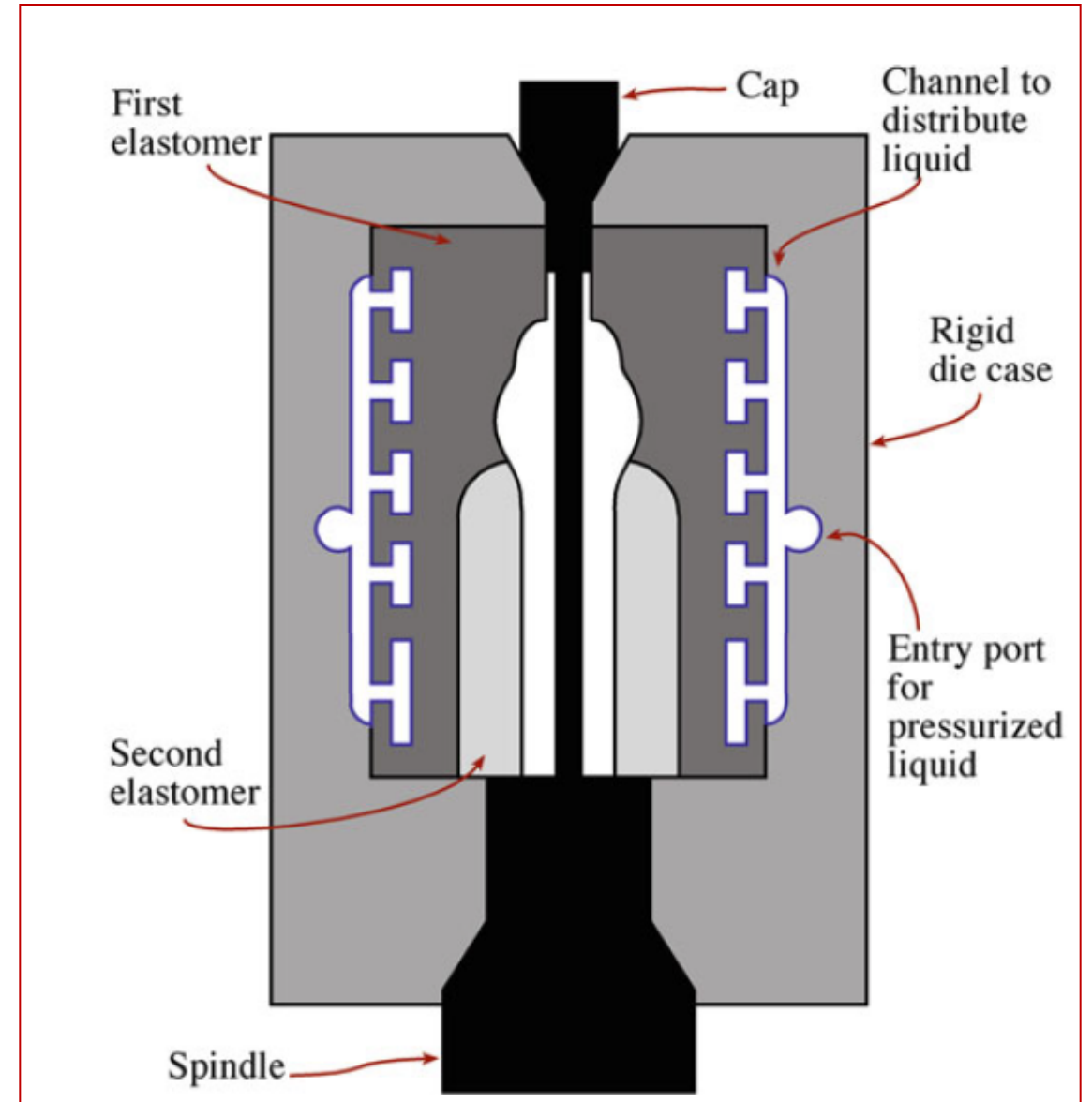
Disadvantages: *Poor shape and dimensional control (particularly for complex shapes)*

Products often require green machining after pressing

Long cycle times (typically 5–60 min) that give low production rates

Dry-bag isostatic pressing

- ❑ to achieve increased production rate and close dimensional tolerances
- ❑ careful positioning and shaping of the fluid channels, use of several different elastomer materials in a single die.
- ❑ parts can be pressed at a rate of 1000–1500 cycles per hour



HOT ISOSTATIC PRESSING

Temperatures can be up to $2,000^{\circ}\text{C}$, pressures are typically in the range 30–100 MPa.

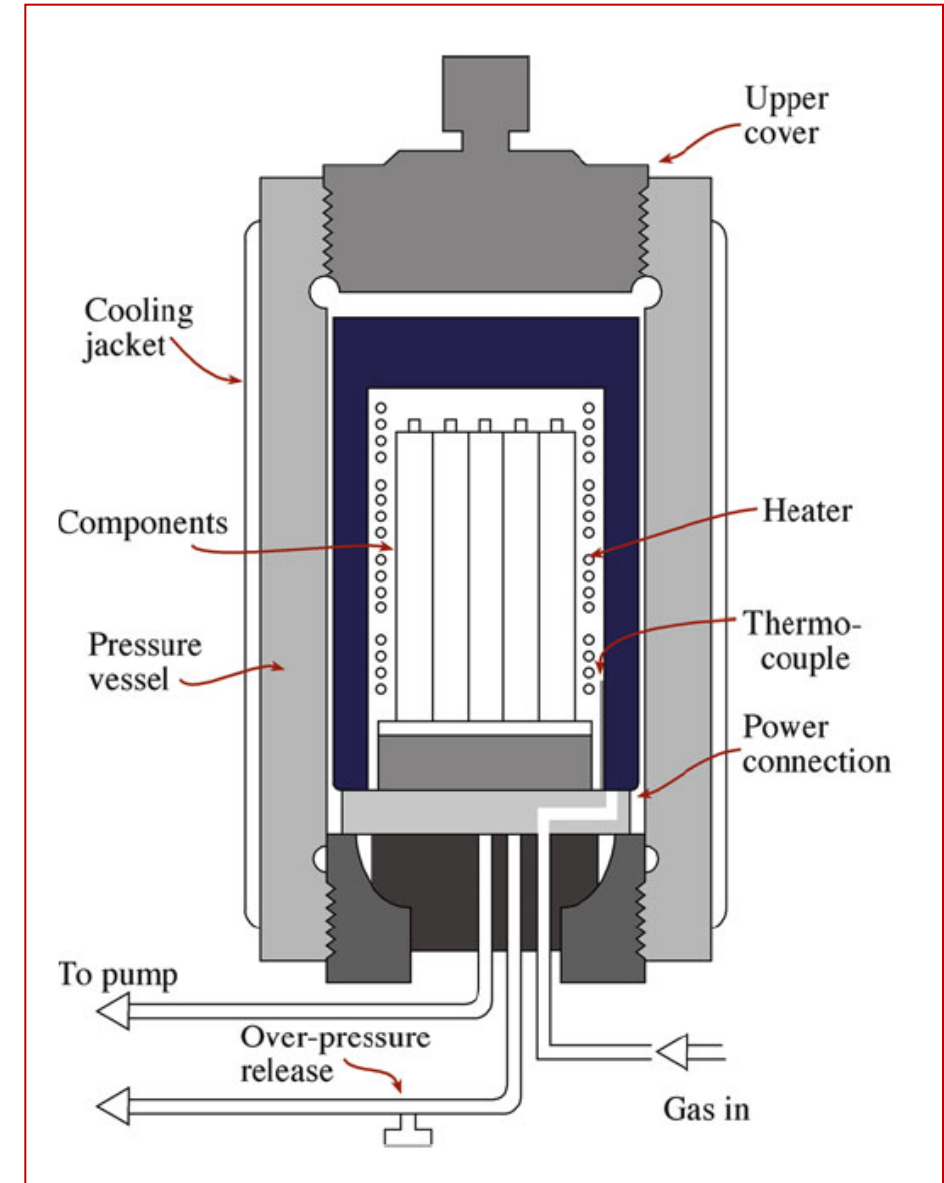
Pressure medium: Argon gas

Nonoxide ceramics can be HIPed to full density while keeping the grain size small and not using additives

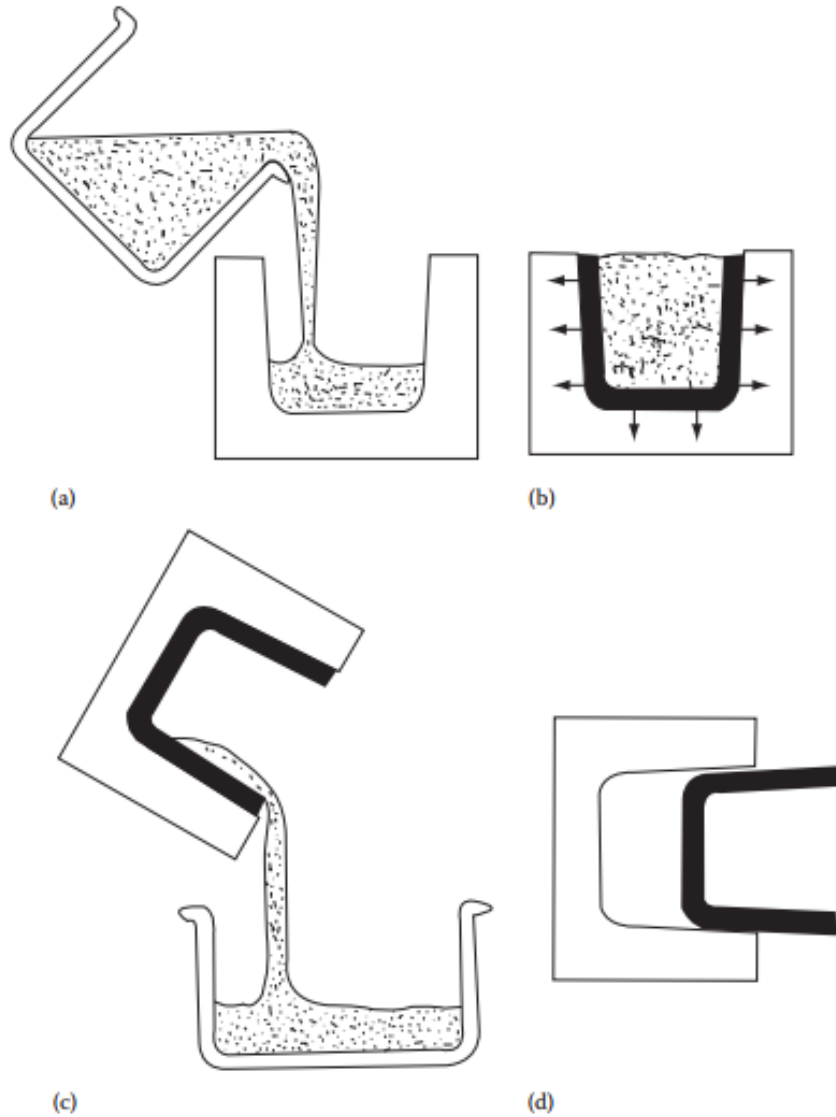
piezoelectric ceramics : BaTiO_3 , SrTiO_3 , lead zirconate titanate (PZT)—for use in acoustic wave filters and oscillators.

Uses: produces dense materials without growing the grains

Disadvantage: cost



SLIP CASTING



Steps involved

Raw materials,
Powder Processing,
slip preparation,
slip casting,
mold removal and drying,
densification

- (a) Fill mold with slip;
- (b) mold extracts liquid and forms compact along mold walls;
- (c) excess slip drained; and
- (d) casting removed after partial drying.

- ❑ It involves ceramic particles suspended in water and cast into porous plaster molds. mold (usually plaster of Paris: $2\text{CaSO}_4\text{H}_2\text{O}$)
- ❑ low-cost way to produce complex shapes.
- ❑ Traditional pottery industry: method for the production of teapots, jugs
- ❑ Large articles, such as wash-hand basins and other whitewares, are also mass-produced by slip casting.
- One limitation with most slip-casting processes is the long time required to cast articles in the mold.
- This results in a large inventory of molds, high labor, and large floor space, all of which add to cost

Technical and structural ceramics:

Alumina crucibles has been successfully used to make complex structural ceramic components such as gas-turbine rotors

Additives for Ceramics

Table 1.5 Common reconsolidation additives

<i>Additive</i>	<i>Function</i>	<i>Shaping operation</i>
Binder	Provide green strength and lubricate	Pressing, slip/tape casting
Lubricant	Decrease die-wall and interparticle friction, assist mould release	Pressing, injection moulding
Compaction aid	Assist particle rearrangement	Pressing
Plasticizer	Control rheology, allow granule plastic deformation	Injection moulding, plastic forming
Deflocculant/ flocculant	pH control, surface charge control	Slip/tape casting
Sintering aid	Assist densification via liquid or solid state	Any

TABLE 22.3 Additives Used in Industrial Pressing Powders

Product	Binder	Plasticizer	Lubricant
Alumina	Polyvinyl alcohol ^a	Polyethylene glycol ^b	Mg Stearate
96% alumina substrates	Polyethylene glycol ^c	None	Talc, ^d clay ^d
Alumina spark plug insulation	Microcrystalline wax emulsion	KOH + tannic acid	Wax, talc ^d clay ^d
MnZn Ferrites	Polyvinyl alcohol ^a	Polyethylene glycol ^b	Zn Stearate
Ba Titanate	Polyvinyl alcohol ^a	Polyethylene glycol ^b	Wax, talc, ^d clay ^d
Steatite	Microcrystalline wax, clay	Water	
Ceramic tile	Clay	Water	Talc, ^d clay ^d
Hotel china	Clay, polysaccharide	Water	Clay ^d
Refractories	Ca/Na lignosulfonate	Water	Stearate